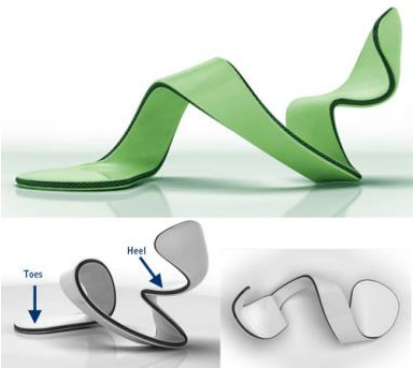
Jules Verne's
(1828-1905)
'NOTILUS'

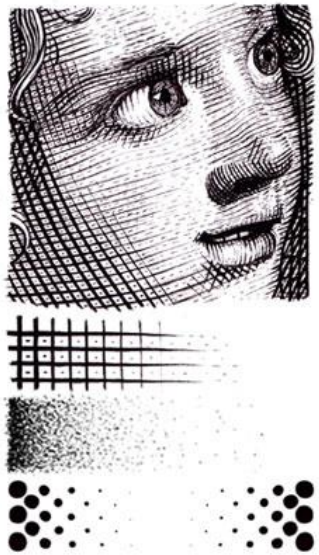
CONCEPT

- **CONCEPT**
- **<u>Creativity evolves around an idea (or Concept)</u>**. The creative idea is linked to general intelligence. It is a mental and neurological process. Concept or idea is perhaps the basic ingredient for any new innovation. Having faced with a problem we search for a solution- concept is born.

  Concept is 'abstract' idea that may be validated through 'information science', cognitive science, metaphysics and philosophy of mind.
- **<u>Concept needs to be validated through scientific reasoning and rational arguments.</u>** Concept has to go through the process of reasoning. For example, 'human being can fly without any device', is a concept that needs to validated through logical mind and reasoning. Once it is validated it would appear as a creative act. Flying above the surface of earth may be a concept; however, it needs logical reasoning verified through various other mechanisms.



How do I make my footwear behave like my natural foot (problem)?



How do I show 3D Form on a flat 2D surface- is a problem. The above drawing shows the technique that would help me to express 3D Volume.

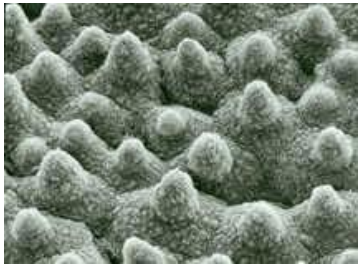
I feel laze to get up and switch off the light (problem).

How do I drain water from the soap case (problem)?

Focused light beam hurts my eyes (problem).

- “**Creativity is a phenomenon whereby something new and somehow valuable is formed**.”

- Creativity (idea) needs to be nurtured and protected from possible damage. If it faces adverse environment there is strong probability it may be killed at its infant stage. Therefore just like an unborn infant needs to be protected in mothers’ womb each idea is unique and needs to be protected till it is matured to face the outer world. The application of creativity can be fostered through education and training. The concept has to be augmented by various mechanism including technology and effective teaching and learning. Every creative product needs constant evaluation and effectiveness with the help of new knowledge and learning.
- The field of creativity includes- idea, scientific theory, musical composition, joke, design etc. It is interdisciplinary in nature that may be verified through various sources.
- Mother Nature is the greatest source of creative ideas. Observation of nature is the most important aspects to inspire and foundation for solutions.
- List as many as possible field of activities that may come under creative act.



Mother-nature has full of surprises awaiting to be discovered.

“The **Velcro**® brand of hook and loop was **invented** by a man named George de Mestral in the 1940's while hunting in the Jura mountains in Switzerland. Mr. de Mestral, a Swiss engineer, realized that the tiny hooks of the cockle-burs were stuck on his pants and in his dog's fur and wondered how they attached themselves.” That gave birth to a creative idea.
(Ref.://www.google.co.in/?gfe_rd=ctrl&ei=IlQgU_6NeTV8ge_34GwCg&gws_rd=cr#q=Invention+of+Velcro)

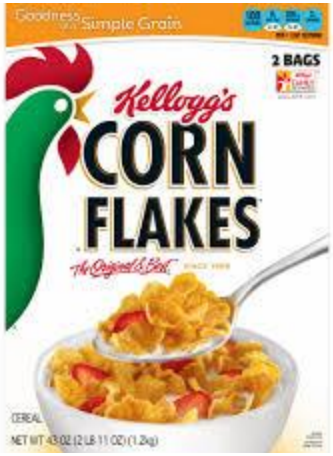
The **lotus effect** refers to self-cleaning properties that are a result of very high water repellence (super-hydrophobicity), as exhibited by the leaves of the lotus flower. Dirt particles are picked up by water droplets due to the micro- and nanoscopic architecture on the surface, which minimizes the droplet's adhesion to that surface. The phenomenon of super-hydrophobicity was first studied by Dettre and Johnson in 1964-using rough hydrophobic surfaces. Their work developed a theoretical model based on experiments with glass beads coated with paraffin.
(http://www.nanowerk.com/spotlight/spotid=19644.php Jan.11, 2016)

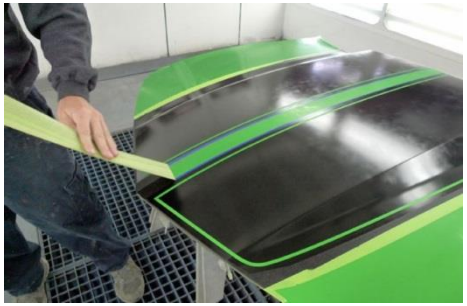
How about designing a device that would clean fabric surface without any water?

- <u>Common understanding of ‘creativity’ is ‘originality’. It is very rare to find anything original</u>. Every new thing that we try to do is always linked some other similar experience. It is almost impossible to think something completely original which has no reference or similarity to anything. Human beings have seen/ experienced seeing avian flying in the sky has inspired to innovating flying machines- Zeppelin, Hot-air Balloon, Airplane, etc. Humans are unable to run faster than many animals, which inspired them to invent various devices which may help us going faster. Nature has abundance of resources to inspire us for innovation. It is important for human beings to search for solutions from nature around us.

Animal runs faster than human being

Man keeps on challenging to go faster





Kellogg Cornflake- "Responsible for making bread dough, one day Kellogg accidentally left his main ingredient -- boiled wheat -- sitting out for several hours. When he came back to roll the ingredient into dough, the wheat became flaky. Curious to see what would happen, Kellogg baked the flaky dough anyway, creating a crunchy and flaky snack. The flakes were a hit with patients, so Kellogg embarked on a mission to enhance the product for large-scale sale."

- **Masking Tape-** In 1923, Richard Drew joined the 3M Company located in St. Paul, Minnesota. At the time, 3M only made sandpaper.
- Drew was product testing 3M's Wetordry brand sandpaper at a local auto body-shop, when he noticed that auto painters were having a hard time making clean dividing lines on two-color paint jobs. Richard Drew was inspired to invent the world's first masking tape in 1925, as a solution to the auto painters' dilemma.

- **Pepsi Cola- "Brad's Drink": Pepsi's Humble Origins**
- Caleb Bradham of New Bern, North Carolina was a **pharmacist**. Like many pharmacists at the turn of the century he had a soda fountain in his drugstore, where he served his customers refreshing drinks that he created himself. His most popular beverage was something he called "Brad's drink," a mix of sugar, water, caramel, lemon oil, kola nuts, nutmeg, and other additives. It became an overnight sensation.

- **Deodorant-** Before the invention of deodorant, people generally battled their offensive smells by masking them with perfumes (a practice dating to the Ancient Egyptians and Greeks). That changed when Mum deodorant came onto the scene in 1888. Unfortunately, we don't actually know whom to thank for saving us all from our stink, as the inventor's name has been lost.
- All we know is that this Philadelphia-based inventor trademarked his invention and distributed it through his nurse under the name of ‘Mum’.

1920s Auto Industry

T- Model Ford Car

- <u>During the time period from 1919 to 1923, primarily North America and parts of Europe experienced the rise of the Roaring Twenties, social, and economic circumstances underwent dramatic changes. The economic power and high employment of the United States allowed Americans to spend more extravagantly on entertainment.</u>

- War veterans returned home seeking relaxation and comfort, instead of returning to their factory or agricultural duties. Watching movies and listening to the newly invented radio became increasingly popular during this period, which further encouraged the desires of people for Hollywood style lives of indulgence and ease. This extravagance was ignited by the introduction of Henry Ford's Model T, a car affectionately known as "Tin Lizzie." Cars became a major source of freedom and greatly altered the standard of living and social patterns, and differentiated suburban and urban living purposes. In addition, the rise of cars led to the creation of new leisure activities and businesses. The car became the center of middle and working class life, until the start of World War II.
- (https://en.wikipedia.org/wiki/Cars_in_the_1920s)